

An overview of Class Diagrams

A Class diagram

- ✓ is a static diagram as it represents the static **view of an application**. Class diagram is not only used for visualizing, describing, and documenting **different aspects of a system** but also for **constructing executable code** of the software application.
- ✓ It **describes** the **attributes and operations of a class** and the constraints imposed on the system. They are widely used in the modelling of object-oriented systems as they are the only UML diagrams which can be mapped directly with object-oriented languages.
- ✓ It shows a collection of **classes, interfaces, associations, collaborations, and constraints**. It is also known as a structural diagram.

Purpose of Class Diagrams

- **Analysis and design** of the static view of an application.
- Describe **responsibilities** of a system.
- **Base for** component and deployment diagrams.
- **Forward and reverse** engineering.

How to Draw a Class Diagram?

Class diagrams have a lot of properties to consider while drawing but here the diagram will be considered from a top level view. As a class diagram graphically represents an application a collection of class diagrams represent the whole system.

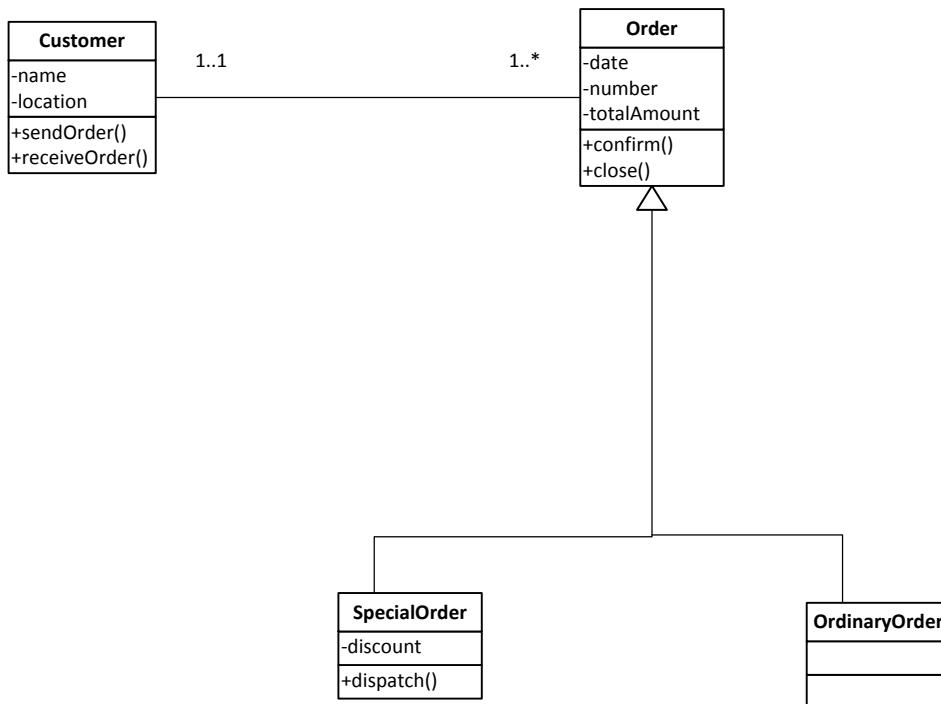
The following points should be remembered while drawing a class diagram –

- The **name of the class** diagram should be meaningful to describe the aspect of the system.
- **Each element and their relationships** should be identified in advance.
- **Responsibility (attributes and methods)** of each class should be clearly identified
- For each class, minimum number of properties should be specified, as unnecessary properties will make the diagram complicated.
- Use notes whenever required to describe some aspect of the diagram. At the end of the drawing it should be understandable to the developer/coder.
- Finally, before making the final version, the diagram should be drawn on plain paper and reworked as many times as possible to make it correct.

The following diagram is an example of an Order System of an application. It describes a particular aspect of the entire application.

- First of all, **Order** and **Customer** are identified as the two elements of the system. They have a **one-to-many relationship** because a customer can have **multiple orders**.
- **Order class is an abstract class** and it has **two concrete classes** (**inheritance relationship**) **SpecialOrder** and **NormalOrder**.
- The two inherited classes have all the properties as the Order class. In addition, they have additional functions like dispatch () and receive ().

The following class diagram has been drawn considering all the points mentioned above.



When to Use Class Diagrams?

A class diagram is used for –

- Describing the static view of the system.
- Showing the collaboration among the elements of the static view.
- Describing the functionalities performed by the system.
- Constructing software applications using object oriented languages.

Adapted from https://www.tutorialspoint.com/uml/uml_class_diagram.htm